TEACHING PORTFOLIO

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This document is a portfolio of my teaching materials. It includes my teaching statement, a syllabus for each course, as well as samples of assignments, notes, evaluations, anonymous feedback, and video recordings.

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1 Teaching statement

In this document, I describe my relevant experience and teaching philosophy. I have taught various courses at the undergraduate and graduate level, mentored high school and undergraduate students, and contributed to curriculum design. I emphasise that studying mathematics is an active and engaging endeavour, through which every student can have deeply meaningful experiences. In each of these roles, I have thus sought to create learning environments in which students feel supported, challenged, and able to succeed.

Currently, I am a J.J. Sylvester Assistant Professor at Johns Hopkins University, where I have taught broadly across the mathematical curriculum during my first two years as an instructor. The courses I have instructed include freshman Calculus I, upper-level undergraduate Calculus on Manifolds and Honors Analysis I, and a two-semester graduate sequence in Partial Differential Equations. My teaching evaluations have been consistently very positive, and I received the student-voted departmental teaching award in both years.

In the classroom I endeavour to maintain an approachable and encouraging presence. Central to this is the guiding principle that every question is worth asking; I want students to be active participants, not spectators, in their own learning. In practice, this means inviting questions, soliciting student responses, and encouraging active engagement with the material. If an example seems counterintuitive, where does our intuition fail? If a result seems too abstract, what examples or applications can make it more concrete?

Introducing concepts through worked examples helps students understand how they might discover a result for themselves, while supporting the development of strong mathematical intuition. I have found this approach especially useful when teaching continuity in both calculus and analysis classes, where the intuitive notion of drawing a curve without lifting one's pen from paper is translated into a formal definition in terms of limits.

To foster wider interest in mathematics, I often emphasise its interconnectedness, showing students how different areas relate to and reinforce one another. This can be particularly effective for students drawn to applications, as it motivates abstract mathematical concepts by presenting them not as isolated definitions, but as tools with significance across the natural sciences. For example, when introducing Lipschitz functions in analysis, I discuss their role in ensuring the existence and uniqueness of solutions to differential equations, which in turn model various physical phenomena.

Careful course preparation and thoughtful assessment design are key components of successful student learning. In my Calculus on Manifolds course, I substantially restructured the material to accommodate varied levels of incoming backgrounds, placing greater emphasis on examples and geometric intuition. I have also used a range of assessment methods alongside traditional written examinations, including quizzes, presentations, group projects, and portfolios. These play valuable roles in providing regular feedback, developing mathematical communication, encouraging collaboration, and allowing students to reflect on their progress. I aim to continue to refine and adopt pedagogical techniques that support my students.

Outside the classroom, I have supervised several reading courses and mentored students on research projects. Mentorship is essential for helping students develop skills that are not always cultivated in the classroom: independence, confidence in their own mathematical judgement, the ability to formulate questions, and the persistence needed to work through uncertainty. My research interests provide a natural source of accessible and meaningful projects that can be tailored to students at different stages of their mathematical development, including those encountering research for the first time; I am enthusiastic about developing further mentoring opportunities.

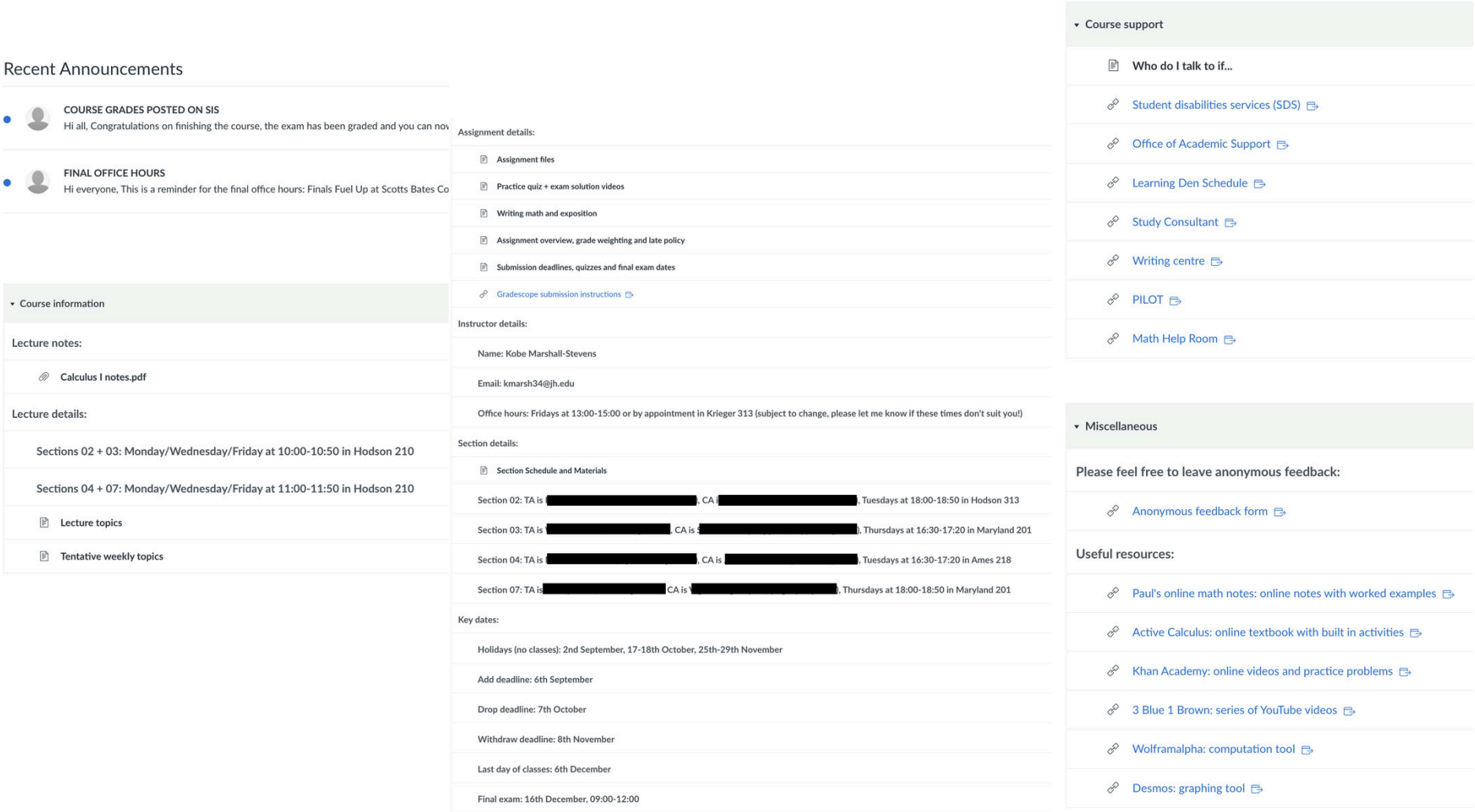
Taken together, my experience and teaching philosophy reflect a commitment to rigorous, supportive environments in which students engage actively. I would be delighted to bring this approach to your department and to contribute to its teaching, mentoring, curriculum development, and wider academic community.

2 Syllabi

In this section I include the syllabus for each course I have instructed, first providing some screenshots of the Canvas template I constructed for Calculus I (and then repeatedly utilised for every other course). I then provide screenshots of each course calendar (with class dates, topics covered, and relevant textbook sections) as well as the assignment overview, grade weighting, and late policy pages.

2.1 Canvas samples

Below are screenshots of my Calculus I Canvas page, similar templates were used for each of the other courses I have instructed.



2.2 Calculus I

Assignment overview: There will be 5 types of assignments for the course:

1. Problem sheets: There will be 12 problem sheets for the course, submitted on Gradescope. These are (with the exception of the first and last) due Monday at 11:59pm EST the week after they have been assigned. Collaboration is encouraged but submissions must be individually written up and submitted.

2. Group projects: There will be 5 group projects for the course, submitted on Gradescope. These are due at Friday 11:59pm EST the week after they have been assigned. Collaboration is, of course, necessary and you will get started on these in your groups during sections and submit these together.

3. Quizzes: There will be 4 written quizzes for the course, given during Friday lectures. Collaboration is prohibited and these will be submitted in the lecture. If you score less than 70% on a quiz you must retake it (on the Thursday of the next week between 7-9pm in Krieger 205). If you score between 70% and 80% you may choose to retake the quiz. Your final quiz grade for each quiz will then be the maximum of the first result and the retake, up to 80%.

4. Personal portfolios: There will be 2 personal portfolios for the course, submitted on Gradescope. These will consist of individual projects, quiz wrappers and an end of course reflection. Collaboration is prohibited, but you are encouraged to ask the course staff (Instructor, TAs, CAs) questions.

5. Final exam: There will be a comprehensive final exam given on 16th December 9am-12pm. More information will be given about this closer to the end of the semester.

Grade weighting: Your final grade will then be calculated based on the following weighting of the above assignments:

- Problem sheets: The 10 highest scoring problem sheets (20% of final grade).
- Group projects: All 5 group projects (10% of final grade).
- Quizzes: All 4 quizzes (40% of final grade).
- Personal portfolios: Both personal portfolios (15% of final grade).
- Final exam: The final exam (15% of final grade).

Late submission policy: There will be a 48 hour late window to upload and/or complete assignments (excluding quizzes and the final exam) and a late penalty of 10% per day applied to the late submissions. No assignments will be accepted after the late deadline has passed except in extenuating circumstances (in the case of extenuating circumstances please coordinate with your case manager and reach out to the instructor and the Head TA as soon as possible).

Week	Date	Topic	Relevant sections of the textbook
1	Monday, August 26	Welcome! Precalculus review, modelling with functions.	1.1, 1.2
	Wednesday, August 28	New functions from old, inverse functions, exponentials and logarithms.	1.3, 1.4, 1.5
	Friday, August 30	Tangent lines, instantaneous velocity, introduction to derivatives.	2.1, 2.7
2	Monday, September 2	*No Classes - Labor Day*	
	Wednesday, September 4	Computing derivatives, derivatives as function, points where functions fail to be differentiable.	2.7, 2.8
	Friday, September 6	Introduction to limits, one-sided limits and infinite limits.	2.2
3	Monday, September 9	Limits laws, types of discontinuity.	2.3, 2.5
	Wednesday, September 11	Continuity, intermediate value theorem (IVT), limits at infinity.	2.5, 2.6
	Friday, September 13	****Quiz 1****	
4	Monday, September 16	Derivatives are continuous, higher derivatives, derivatives of polynomials and exponentials.	2.8, 3.1
	Wednesday, September 18	Product and quotient rule, derivatives of trigonometric functions.	3.2, 3.3
	Friday, September 20	Chain rule.	3.4
5	Monday, September 23	Implicit differentiation, inverse derivatives (for logarithms and inverse trigonometric functions).	3.5, 3.6
	Wednesday, September 25	Maximum/minimum values, extreme value theorem (EVT), critical points/values, closed interval method.	4.1
	Friday, September 27	Rolle's theorem, mean value theorem (MVT) and its applications.	4.2
6	Monday, September 30	How derivatives affect the shape of a graph, first and second derivative test, curve sketching.	4.3, 4.5
	Wednesday, October 2	More curve sketching, rates of change in science, exponential growth and decay.	4.5, 3.7, 3.8
	Friday, October 4	****Quiz 2****	
7	Monday, October 7	Logarithmic differentiation (not examinable), related rates.	3.6, 3.9
	Wednesday, October 9	More related rates, linear approximation (not examinable).	3.9, 3.10
	Friday, October 11	Indeterminate forms and L'Hospital's rule.	4.4
8	Monday, October 14	More indeterminate forms, optimisation problems.	4.4, 4.7
	Wednesday, October 16	More optimisation problems, review of limits and derivatives.	4.7
	Friday, October 18	*No Classes - Fall Break*	
9	Monday, October 21	Antiderivatives.	4.9
	Wednesday, October 23	Areas and distances, Riemann sums.	5.1
	Friday, October 25	Introduction to the definite integral, signed area interpretation.	5.2
10	Monday, October 28	Properties of the definite integral.	5.2
	Wednesday, October 30	Functions that are not integrable, the definite integral as a function.	5.3
	Friday, November 1	****Quiz 3****	
11	Monday, November 4	The fundamental theorem of calculus (FTC).	5.3
	Wednesday, November 6	Evaluating integrals using the fundamental theorem of calculus.	5.3
	Friday, November 8	The indefinite integral and the net change theorem.	5.4
12	Monday, November 11	Integration by substitution (a.k.a. u-substitution).	5.5
	Wednesday, November 13	More integration by substitution and quiz preparation.	5.5
	Friday, November 15	****Quiz 4****	
13	Monday, November 18	Areas between curves.	6.1
	Wednesday, November 20	Integration by parts.	7.1
	Friday, November 22	Exam information, course reflection and Q&A.	
14	November 25-29	*No Classes - Thanksgiving Break*	
15	Monday, December 2	Exam preparation and Q&A.	
	Wednesday, December 4	Exam preparation and Q&A.	
	Friday, December 6	Exam preparation and Q&A.	

2.3 Calculus on Manifolds

Assignment overview: There will be 2 types of assignments for the course:

- Homework: There will be 4 homework sets for the course, submitted on Gradescope. Collaboration is encouraged but submissions must be individually written up and submitted.
- Quiz: There will be 3 quizzes for the course, completed in class.

Grade weighting: Your final grade will be calculated based on the following weighting of the above assignments:

- Homework: All 4 homework sets (each 15% of final grade).
- Quiz: All 3 quizzes (first worth 10%, second and third worth 15% of final grade).

Late submission policy: No submissions will be accepted after the submission deadline has passed except in extenuating circumstances (in the case of extenuating circumstances please reach out to the instructor as soon as possible).

Week	Date	Lecture Topic	Relevant Chapters of Lee's book
1	Tuesday, January 21st	Motivation and examples of manifolds.	1
	Thursday, January 23rd	A primer in topology, defining topological manifolds.	1, Appendix A
2	Tuesday, January 28th	Charts & atlases, transition maps, the definition and examples of smooth manifolds.	1, Appendix C
	Thursday, January 30th	Smooth maps on and between manifolds, diffeomorphisms, manifolds as quotient spaces.	2, 21
3	Tuesday, February 4th	Partitions of unity and the abundance of smooth functions on manifolds.	2
	Thursday, February 6th	**Quiz 1**	
4	Tuesday, February 11th	Tangent vectors and spaces via derivations.	3
	Thursday, February 13th	The differential, dimension of the tangent space, and coordinate calculations *(homework 1 due).*	3
5	Tuesday, February 18th	Inverse function theorem for manifolds, maximal rank and regular value theorems.	4
	Thursday, February 20th	Immersions, submersions, and embeddings: submanifolds and the Whitney embedding theorems.	4, 6
6	Tuesday, February 25th	The tangent bundle, global differential, and vector fields.	3, 8
	Thursday, February 27th	Cotangent bundle, covector fields, and the differential	11
7	Tuesday, March 4th	More on the differential, pullbacks, and the line integral.	11
	Thursday, March 6th	Vector bundles, sections, and parallelisability *(homework 2 due).*	10
8	Tuesday, March 11th	A primer in multilinear algebra: tensors, symmetric and alternating tensors, and the wedge product.	12, Appendix B
	Thursday, March 13th	**Quiz 2**	
9	Tuesday, March 18th	*Spring break, no lecture*	
	Thursday, March 20th	*Spring break, no lecture*	
10	Tuesday, March 25th	Riemannian metrics and differential forms.	12, 13, 14
	Thursday, March 27th	Orientations and orientation forms/atlases.	15
11	Tuesday, April 1st	Integration of forms.	16
	Thursday, April 3rd	Manifolds with boundary.	1, 15
12	Tuesday, April 8th	The exterior derivative on forms (*homework 3 due).*	14
	Thursday, April 10th	Stokes' theorem.	16
13	Tuesday, April 15th	Applications of Stokes' theorem.	16
	Thursday, April 17th	Closed and exact forms, de Rham cohomology.	17
14	Tuesday, April 22nd	Course review & connections to other topics *(homework 4 due).*	
	Thursday, April 24th	**Quiz 3**	

2.4 Honors Analysis I

Assignment overview: There will be 2 types of assignments for the course:

- Homework: There will be 10 homeworks for the course, submitted on Gradescope. Collaboration is encouraged but submissions must be individually written up and submitted.
- Quiz: There will be 2 quizzes for the course, completed in class.

Grade weighting: Your final grade will be calculated based on the following weighting of the above assignments:

- Homework: All 10 homeworks (each 6% of final grade).
- Quiz: Both quizzes (each 20% of final grade).

Late submission policy: No submissions will be accepted after the submission deadline has passed except in extenuating circumstances (in the case of extenuating circumstances please reach out to the instructor as soon as possible).

Week	Date	Lecture Topic	Relevant Chapter(s) of Rudin's book
1	August 25th	Introduction, ordered sets, supremum and infimum.	1
	August 27th	Fields, the real numbers, and roots.	1
2	September 1st	*Labour day, no lecture.*	
	September 3rd	Extended reals, complex numbers, Euclidean space, the triangle and Cauchy--Schwartz inequality	1
3	September 8th	Finite, countable, and uncountable sets	2
	September 10th (Homework 1 due)	Cantor's diagonalisation argument, metric spaces, open and closed sets	2
4	September 15th	More on open and closed sets, the closure of a set	2
	September 17th (Homework 2 due)	Relative openness and compactness	2
5	September 22nd	More on compactness and the Heine--Borel theorem	2
	September 24th (Homework 3 due)	Perfect sets, the uncountability of the real line, and the Cantor set	2
6	September 29th	Connectedness	2
	October 1st (Homework 4 due)	Limits of sequences, the limit laws	3
7	October 6th	Subsequences, Cauchy sequences, and completeness	3
	October 8th (Homework 5 due)	Monotone sequences, limsup and liminf	3
8	October 13th	Series	3
	October 15th	**Quiz 1**	
9	October 20th	Continuity	4
	October 22nd (Homework 6 due)	The extreme value theorem and uniform continuity	4
10	October 27th	Continuity and connectedness, discontinuities, and monotone functions	4
	October 29th (Homework 7 due)	Normed vector spaces	7
11	November 3rd	Differentiation	5
	November 5th (Homework 8 due)	Darboux's and Taylor's theorem, multivariable differentiation	5
12	November 10th	Sequences and series of functions, uniform continuity	7
	November 12th (Homework 9 due)	More on uniform continuity and function spaces	7
13	November 17th	A continuous but nowhere differentiable function, equicontinuity	7
	November 19th (Homework 10 due)	The Arzelà--Ascoli theorem	7
14	November 24th	*Fall recess, no lecture.*	
	November 26th	*Fall recess, no lecture.*	
15	December 1st	Construction of the real numbers and equivalence of the completeness axioms (non-examinable)	1 - Appendix
	December 3rd	**Quiz 2**	

2.5 Partial Differential Equations I

Assignment overview: There will be 2 types of assignments for the course:

- Homework: There will be 4 homeworks for the course, submitted on Gradescope. Collaboration is encouraged but submissions must be individually written up and submitted.
- Quiz: There will be 2 quizzes for the course, completed in class.

Grade weighting: Your final grade will be calculated based on the following weighting of the above assignments:

- Homework: All 4 homeworks (each 15% of final grade).
- Quiz: Both quizzes (each 20% of final grade).

Late submission policy: No submissions will be accepted after the submission deadline has passed except in extenuating circumstances (in the case of extenuating circumstances please reach out to the instructor as soon as possible).

Week	Date	Lecture Topic	Relevant Sections of Evans' book
1	August 25th	Introduction, function spaces	1, A, B
	August 27th	Functional analysis preliminaries, course outline	D.1, D.2, D.3
2	September 1st	*Labour day, no lecture.*	
	September 3rd	Weak derivatives	5.2.1
3	September 8th	Sobolev spaces, absolute continuity and one dimensional Sobolev spaces	5.2.2, 5.2.3
	September 10th	Mollification, interior approximation of Sobolev functions	5.3.1, 5.3.2, C.4, E.4
4	September 15th	Global approximation of Sobolev functions	5.3.3
	September 17th	Extensions of Sobolev functions	5.4
5	September 22nd	The trace operator	5.5
	September 24th (Homework 1 due)	The Gagliardo--Nirenberg--Sobolev inequality, the isoperimetric inequality	5.6.1
6	September 29th	Hölder spaces, the Morrey inequality	5.6.2
	October 1st	Embeddings of Sobolev spaces and compactness	5.6.3, 5.7, C.7
7	October 6th	Rellich--Kondrachov compactness	5.7
	October 8th	Weak compactness and the direct method of the calculus of variations	D.4
8	October 13th	Difference quotients, Lipschitz functions, differentiability almost everywhere	5.8.2, 5.8.3, E.4
	October 15th	**Quiz 1**	
9	October 20th	Second order linear PDE, H^{-1}, weak solutions	5.9.1, 6.1.1, 6.1.2
	October 22nd (Homework 2 due)	Ellipticity, boundedness and coercivity of the associated bilinear form	6.1.1, 6.2.2
10	October 27th	The Lax--Milgram theorem, the first existence theorem for elliptic PDE	6.2.1, 6.2.2
	October 29th	An elliptic PDE with no existence or uniqueness, Fredholm theory	D.2, D.5
11	November 3rd	The Fredholm alternative, the adjoint operator, the second existence theorem for elliptic PDE	6.2.3, D.5
	November 5th (Homework 3 due)	Spectrum and eigenvalues of elliptic operators, the third existence theorem for elliptic PDE	6.2.3, D.5
12	November 10th	Regularity theory for weak solutions, interior regularity	6.3, 6.3.1
	November 12th	Higher interior regularity, bootstrapping	6.3.1
13	November 17th	Boundary regularity	6.3.2
	November 19th (Homework 4 due)	Existence for a nonlinear elliptic PDE	8.2
14	November 24th	*Fall recess, no lecture.*	
	November 26th	*Fall recess, no lecture.*	
15	December 1st	Regularity via bootstrapping for a nonlinear elliptic PDE (as motivation for PDE II)	8.3
	December 3rd	**Quiz 2**	

2.6 Partial Differential Equations II

Assignment overview: There will be 1 type of assignment for the course:

- Presentations: Each student will give a short (exact timing to be determined) presentation on a topic agreed with the instructor.

Grade weighting: Your final grade will be calculated based on the following weighting of the above assignments and attendance:

- Presentations: Worth 50% of the grade.
- Attendance: Worth 50% of the grade.

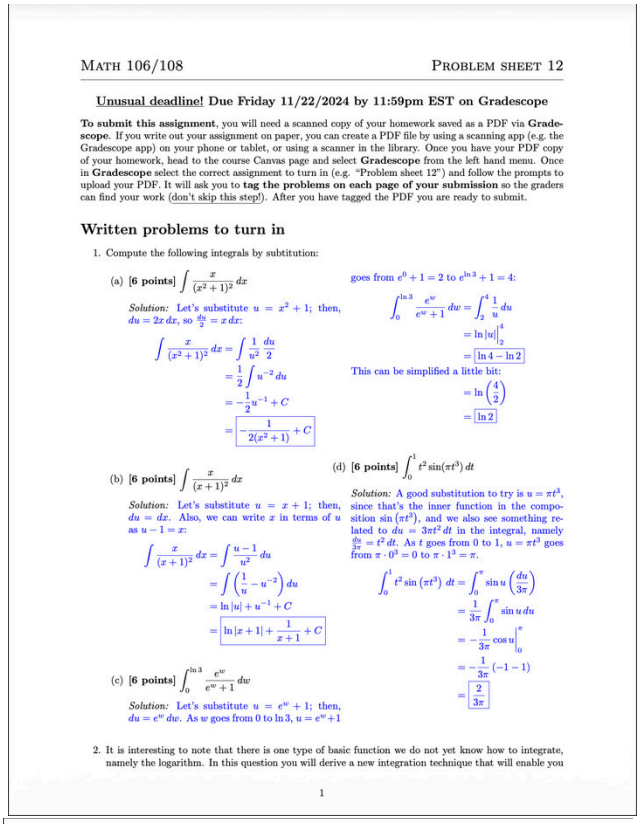
Late submission policy: No submissions will be accepted after the submission deadline has passed except in extenuating circumstances (in the case of extenuating circumstances please reach out to the instructor as soon as possible).

Week	Date	Lecture Topic	References
1	January 19th	*MLK jr. day, no lecture*	
	January 21st	Introduction to Hilbert's XIXth problem, strategy	Regularity Theory for Elliptic PDE, Chapter 3
2	January 26th	Existence and uniqueness of minimisers	Regularity Theory for Elliptic PDE, Chapter 3
	January 28th	Di Giorgi--Nash, sub/super-solutions, the Caccioppoli inequality	Regularity Theory for Elliptic PDE, Chapter 3
3	February 2nd	The L^2 to L^∞ estimate, alternate approach via Moser iteration	Regularity Theory for Elliptic PDE, Chapter 3
	February 4th	Oscillation decay implies Hölder continuity, De Giorgi's isoperimetric inequality, quantitative jumping	Regularity Theory for Elliptic PDE, Chapter 3
4	February 9th	The L^∞ to $C^{0,\alpha}$ estimate, minimisers are $C^{1,\alpha}$, further results	Regularity Theory for Elliptic PDE, Chapter 3
	February 11th	Properties of harmonic functions	Gilbarg and Trudinger, Chapter 2
5	February 16th	More on harmonic functions	Gilbarg and Trudinger, Chapter 2
	February 18th	General non-divergence form elliptic operators, strict and weak maximum principles	Gilbarg and Trudinger, Chapter 3
6	February 23rd	Uniqueness for the Dirichlet problem, comparison principles, Hopf boundary point lemma	Gilbarg and Trudinger, Chapter 3
	February 25th	Strong maximum principle, uniqueness for the Neumann problem, maximum principle a priori estimates	Gilbarg and Trudinger, Chapter 3
7	March 2nd	Introduction to Schauder estimates via informal ideas and examples, Hölder space recap	Simon's Schauder estimates by scaling/Gilbarg and Trudinger, Chapter 6
	March 4th	Interior Schauder estimates for non-divergence form, reduction to "baby Schauder" via Simon's absorbing lemma	Simon's Schauder estimates by scaling/Gilbarg and Trudinger, Chapter 6
8	March 9th	Schauder estimates by scaling	Simon's Schauder estimates by scaling/Gilbarg and Trudinger, Chapter 6
	March 11th	Higher interior Schauder estimates, Schauder estimates for divergence form, completion of Hilbert's XIXth problem	Simon's Schauder estimates by scaling/Regularity Theory for Elliptic PDE, Chapter 3
9	March 16th	*Spring break, no lecture*	
	March 18th	*Spring break, no lecture*	
10	March 23rd	Boundary and global Schauder estimates	Simon's Schauder estimates by scaling/Gilbarg and Trudinger, Chapter 6
	March 25th	The continuity method	Gilbarg and Trudinger, Chapter 6
11	March 30th	Unique solvability of the $C^{2,\alpha}$ Dirichlet problem on balls, continuous sub/super-solutions	Gilbarg and Trudinger, Chapter 6
	April 1st	Lifts of subsolutions, subfunctions, interior Perron	Gilbarg and Trudinger, Chapter 6
12	April 6th	Boundary continuity of the Perron solution by barriers	Gilbarg and Trudinger, Chapter 6
	April 8th	Fredholm alternative for non-divergence form, higher regularity of solutions, application to the MSE	Gilbarg and Trudinger, Chapter 6
13	April 13th	Leray--Schauder fixed point theorem, existence for the quasilinear Dirichlet problem via a priori $C^{1,\beta}$ estimates	Gilbarg and Trudinger, Chapter 11
	April 15th	Application of the theory to the MSE	Gilbarg and Trudinger, Chapter 13
14	April 20th	**Presentations**	
	April 22nd	**Presentations**	
15	April 27th	**Presentations**	

3 Assignment samples

3.1 Problem sheets

Screenshots of a single page of a problem sheet (two with typed solutions in blue) for the courses Calculus I, Calculus on Manifolds, Honors Analysis I, and Partial Differential Equations I are included below.



MATH 106/108 PROBLEM SHEET 12

Unusual deadline! Due Friday 11/22/2024 by 11:59pm EST on Gradescope

To submit this assignment, you will need a scanned copy of your homework saved as a PDF via **Gradescope**. If you write out your assignment on paper, you can create a PDF file by using a scanning app (e.g. the Gradescope app) on your phone or tablet, or using a scanner in the library. Once you have your PDF copy of your homework, head to the course Canvas page and select **Gradescope** from the left hand menu. Once in **Gradescope** select the correct assignment to turn in (e.g. "Problem sheet 12") and follow the prompts to upload your PDF. It will ask you to **tag the problems on each page of your submission** so the graders can find your work (don't skip this step!). After you have tagged the PDF you are ready to submit.

Written problems to turn in

1. Compute the following integrals by subtitution:

(a) **[6 points]** $\int \frac{x}{(x^2+1)^2}\,dx$

Solution: Let's substitute $u = x^2+1$; then, $du = 2x\,dx$, so $\frac{du}{2} = x\,dx$:

$$\int \frac{x}{(x^2+1)^2}\,dx = \int \frac{1}{u^2}\frac{du}{2} = \frac{1}{2}\int u^{-2}\,du = -\frac{1}{2}u^{-1} + C = \boxed{-\frac{1}{2(x^2+1)} + C}$$

(b) **[6 points]** $\int \frac{x}{(x+1)^2}\,dx$

Solution: Let's substitute $u = x+1$; then, $du = dx$. Also, we can write x in terms of u as $u - 1 = x$:

$$\int \frac{x}{(x+1)^2}\,dx = \int \frac{u-1}{u^2}\,du = \int \left(\frac{1}{u} - u^{-2}\right)du = \ln|u| + u^{-1} + C = \boxed{\ln|x+1| + \frac{1}{x+1} + C}$$

(c) **[6 points]** $\int_0^{\ln 3} \frac{e^w}{e^w+1}\,dw$

Solution: Let's substitute $u = e^w + 1$; then, $du = e^w\,dw$. As w goes from 0 to $\ln 3$, $u = e^w + 1$ goes from $e^0 + 1 = 2$ to $e^{\ln 3} + 1 = 4$:

$$\int_0^{\ln 3} \frac{e^w}{e^w+1}\,dw = \int_2^4 \frac{1}{u}\,du = \ln|u|\Big|_2^4 = \boxed{\ln 4 - \ln 2}$$

This can be simplified a little bit:

$$= \ln\left(\frac{4}{2}\right) = \boxed{\ln 2}$$

(d) **[6 points]** $\int_0^1 t^2 \sin(\pi t^3)\,dt$

Solution: A good substitution to try is $u = \pi t^3$, since that's the inner function in the composition $\sin(\pi t^3)$, and we also see something related to $du = 3\pi t^2\,dt$ in the integral, namely $\frac{du}{3\pi} = t^2\,dt$. As t goes from 0 to 1, $u = \pi t^3$ goes from $\pi \cdot 0^3 = 0$ to $\pi \cdot 1^3 = \pi$.

$$\int_0^1 t^2 \sin(\pi t^3)\,dt = \int_0^\pi \sin u \left(\frac{du}{3\pi}\right) = \frac{1}{3\pi}\int_0^\pi \sin u\,du = -\frac{1}{3\pi}\cos u\Big|_0^\pi = -\frac{1}{3\pi}(-1) = \boxed{\frac{2}{3\pi}}$$

2. It is interesting to note that there is one type of basic function we do not yet know how to integrate, namely the logarithm. In this question you will derive a new integration technique that will enable you

1



NAME: HOPKINS ID:

Quiz 2, Calculus on Manifolds, Spring 2025

JHU Ethics Policy reminder: Each member of the University bears a personal responsibility to uphold the ethical standards of the institution. You should not discuss the content of this quiz with other students in the course until everyone has taken it. Please **sign below** to indicate that you understand the ethics implications in this quiz environment:

You have 60 minutes to answer the following 5 questions:

1. **[5 points]** Define the tangent space at a point on a manifold and state its dimension.

Solution: Let $x \in M$ be a point in a manifold. The **tangent space** to M at x, denoted T_xM, is the collection of all **tangent vectors**, which we define as linear functionals $D : C^\infty(M) \to \mathbb{R}$ with the property that

$$D(fg) = g(x)D(f) + f(x)D(g)$$

for each $f, g \in C^\infty(M)$. The dimension of the tangent space is the same as the manifold.

2. **[10 points]** If $M \subset N$ is a submanifold and $x \in M$, show that $T_xM \subset T_xN$.

Solution: Every tangent vector arises as the velocity of some curve (as shown in homework 2), thus as every curve in M is also a curve in N the tangent vectors to M at a point x must also be tangent vectors to N at x. Alternatively one can use the fact that the inclusion map $\iota : M \to N$ is smooth and diffeomorphic onto its image (as it is an embedding of M into N), hence its differential $d_x\iota : T_xM \to T_xN$ is injective and provides an inclusion of vector spaces.

3. **[10 points]** Is the cross, $(\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}) \subset \mathbb{R}^2$, a manifold? Justify your answer.

Solution: Notice that away from the origin in $\mathbb{R}^2$ the pieces of the cross are a 1-manifold (as it consists of four half rays diffeomorphic to the open interval $(0, \infty)$); thus if the cross were a manifold it would be 1-dimensional. On the other hand, consider the curves $(t, 0)$ and $(0, t)$ for each $t \in \mathbb{R}$. These produce tangent vectors, $(1, 0)$ and $(0, 1)$, at the origin that span $\mathbb{R}^2$; thus the tangent space at the origin, if it existed, would be 2-dimensional. This contradicts the fact that the tangent space should be the same as that of the manifold (which should be 1-dimensional).

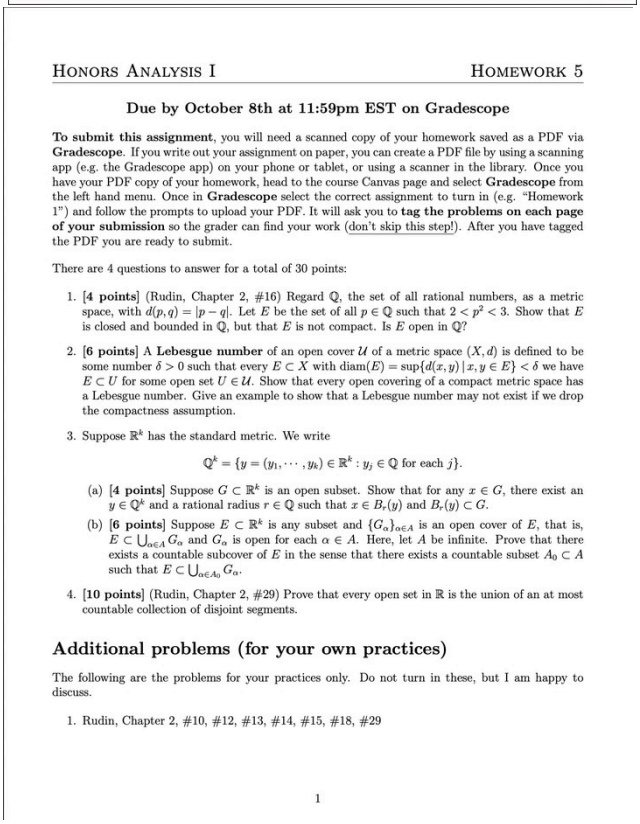
4. Let $\pi : S^n \subset \mathbb{R}^{n+1} \to \mathbb{R}$ be the projection, $\pi(x_1, \ldots, x_{n+1}) = x_{n+1}$.

(i) **[10 points]** Find, by means of the regular value theorem, the values of $y \in \mathbb{R}$ for which $\pi^{-1}(y)$ is an $(n-1)$-manifold. What are the other non-empty $\pi^{-1}(y)$?

(ii) **[10 points]** For each $y \in \mathbb{R}$ such that $\pi^{-1}(y)$ is an $(n-1)$-manifold, compute the tangent space, $T_x\pi^{-1}(y)$, at each $x \in \pi^{-1}(y)$.

(iii) **[5 points]** Geometrically, what do each of the non-empty $\pi^{-1}(y)$ correspond to? (hint: draw a picture when $n = 2$). Guess which manifolds they are diffeomorphic to.

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HONORS ANALYSIS I HOMEWORK 5

Due by October 8th at 11:59pm EST on Gradescope

To submit this assignment, you will need a scanned copy of your homework saved as a PDF via **Gradescope**. If you write out your assignment on paper, you can create a PDF file by using a scanning app (e.g. the Gradescope app) on your phone or tablet, or using a scanner in the library. Once you have your PDF copy of your homework, head to the course Canvas page and select **Gradescope** from the left hand menu. Once in **Gradescope** select the correct assignment to turn in (e.g. "Homework 1") and follow the prompts to upload your PDF. It will ask you to **tag the problems on each page of your submission** so the grader can find your work (don't skip this step!). After you have tagged the PDF you are ready to submit.

There are 4 questions to answer for a total of 30 points:

1. **[4 points]** (Rudin, Chapter 2, #16) Regard $\mathbb{Q}$, the set of all rational numbers, as a metric space, with $d(p, q) = |p - q|$. Let E be the set of all $p \in \mathbb{Q}$ such that $2 < p^2 < 3$. Show that E is closed and bounded in $\mathbb{Q}$, but that E is not compact. Is E open in $\mathbb{Q}$?

2. **[6 points]** A **Lebesgue number** of an open cover $\mathcal{U}$ of a metric space (X, d) is defined to be some number $\delta > 0$ such that every $E \subset X$ with $\operatorname{diam}(E) = \sup\{d(x, y) \mid x, y \in E\} < \delta$ we have $E \subset U$ for some open set $U \in \mathcal{U}$. Show that every open covering of a compact metric space has a Lebesgue number. Give an example to show that a Lebesgue number may not exist if we drop the compactness assumption.

3. Suppose $\mathbb{R}^k$ has the standard metric. We write

$$\mathbb{Q}^k = \{y = (y_1, \cdots, y_k) \in \mathbb{R}^k : y_j \in \mathbb{Q} \text{ for each } j\}.$$

(a) **[4 points]** Suppose $G \subset \mathbb{R}^k$ is an open subset. Show that for any $x \in G$, there exist an $y \in \mathbb{Q}^k$ and a rational radius $r \in \mathbb{Q}$ such that $x \in B_r(y)$ and $B_r(y) \subset G$.

(b) **[6 points]** Suppose $E \subset \mathbb{R}^k$ is any subset and $\{G_\alpha\}_{\alpha \in A}$ is an open cover of E, that is, $E \subset \bigcup_{\alpha \in A} G_\alpha$ and G_α is open for each $\alpha \in A$. Here, let A be infinite. Prove that there exists a countable subcover of E in the sense that there exists a countable subset $A_0 \subset A$ such that $E \subset \bigcup_{\alpha \in A_0} G_\alpha$.

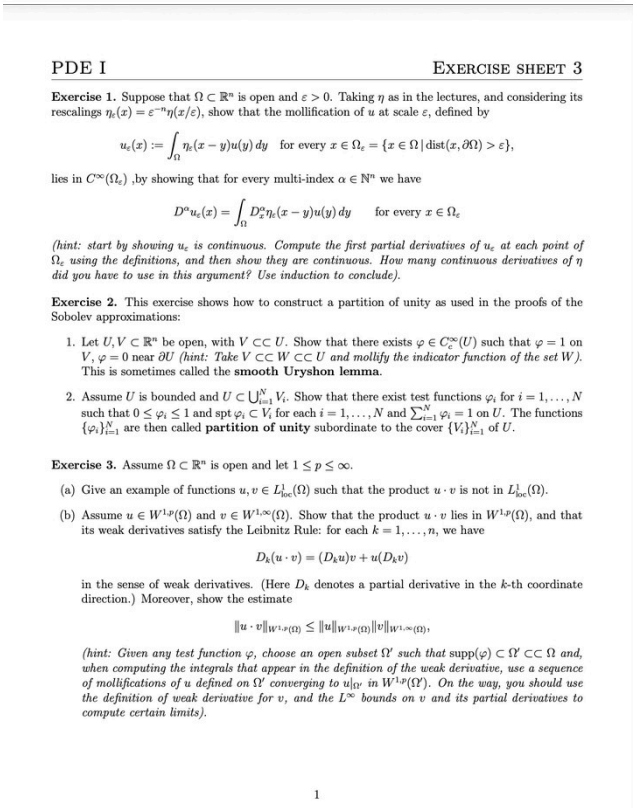
4. **[10 points]** (Rudin, Chapter 2, #29) Prove that every open set in $\mathbb{R}$ is the union of an at most countable collection of disjoint segments.

Additional problems (for your own practices)

The following are the problems for your practices only. Do not turn in these, but I am happy to discuss.

1. Rudin, Chapter 2, #10, #12, #13, #14, #15, #18, #29

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PDE I EXERCISE SHEET 3

Exercise 1. Suppose that $\Omega \subset \mathbb{R}^n$ is open and $\varepsilon > 0$. Taking η as in the lectures, and considering its rescalings $\eta_\varepsilon(x) = \varepsilon^{-n}\eta(x/\varepsilon)$, show that the mollification of u at scale ε, defined by

$$u_\varepsilon(x) := \int_\Omega \eta_\varepsilon(x - y)u(y)\,dy \quad \text{for every } x \in \Omega_\varepsilon = \{x \in \Omega \mid \operatorname{dist}(x, \partial\Omega) > \varepsilon\},$$

lies in $C^\infty(\Omega_\varepsilon)$,by showing that for every multi-index $\alpha \in \mathbb{N}^n$ we have

$$D^\alpha u_\varepsilon(x) = \int_\Omega D^\alpha_x \eta_\varepsilon(x - y)u(y)\,dy \qquad \text{for every } x \in \Omega_\varepsilon$$

(hint: start by showing u_ε is continuous. Compute the first partial derivatives of u_ε at each point of Ω_ε using the definitions, and then show they are continuous. How many continuous derivatives of η did you have to use in this argument? Use induction to conclude).

Exercise 2. This exercise shows how to construct a partition of unity as used in the proofs of the Sobolev approximations:

1. Let $U, V \subset \mathbb{R}^n$ be open, with $V \subset\subset U$. Show that there exists $\varphi \in C_c^\infty(U)$ such that $\varphi = 1$ on V, $\varphi = 0$ near ∂U *(hint: Take $V \subset\subset W \subset\subset U$ and mollify the indicator function of the set W).* This is sometimes called the **smooth Uryshon lemma**.

2. Assume U is bounded and $U \subset \bigcup_{i=1}^N V_i$. Show that there exist test functions φ_i for $i = 1, \ldots, N$ such that $0 \le \varphi_i \le 1$ and $\operatorname{spt}\varphi_i \subset V_i$ for each $i = 1, \ldots, N$ and $\sum_{i=1}^N \varphi_i = 1$ on U. The functions $\{\varphi_i\}_{i=1}^N$ are then called **partition of unity** subordinate to the cover $\{V_i\}_{i=1}^N$ of U.

Exercise 3. Assume $\Omega \subset \mathbb{R}^n$ is open and let $1 \le p \le \infty$.

(a) Give an example of functions $u, v \in L^1_{\text{loc}}(\Omega)$ such that the product $u \cdot v$ is not in $L^1_{\text{loc}}(\Omega)$.

(b) Assume $u \in W^{1,p}(\Omega)$ and $v \in W^{1,\infty}(\Omega)$. Show that the product $u \cdot v$ lies in $W^{1,p}(\Omega)$, and that its weak derivatives satisfy the Leibnitz Rule: for each $k = 1, \ldots, n$, we have

$$D_k(u \cdot v) = (D_k u)v + u(D_k v)$$

in the sense of weak derivatives. (Here D_k denotes a partial derivative in the k-th coordinate direction.) Moreover, show the estimate

$$\|u \cdot v\|_{W^{1,p}(\Omega)} \le \|u\|_{W^{1,p}(\Omega)}\|v\|_{W^{1,\infty}(\Omega)},$$

(hint: Given any test function φ, choose an open subset Ω' such that $\operatorname{supp}(\varphi) \subset \Omega' \subset\subset \Omega$ and, when computing the integrals that appear in the definition of the weak derivative, use a sequence of mollifications of u defined on Ω' converging to $u|_{\Omega'}$ in $W^{1,p}(\Omega')$. On the way, you should use the definition of weak derivative for v, and the L^∞ bounds on v and its partial derivatives to compute certain limits).

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3.2 Quizzes/Exams

Screenshots of a single page of a quiz or exam (one with typed solutions in blue) for the courses Calculus I, Calculus on Manifolds, Honors Analysis I, and Partial Differential Equations I are included below.

NAME: HOPKINS ID:

5. Determine whether the following statements are true or false (circle your answers). You do not need to justify your answer.

(a) **[2 points]** There is a function that is differentiable but not continuous. True False

(b) **[2 points]** $\frac{d^6}{dx^6}(e^{2x}+\sin(x)) = 64e^{2x}+\sin(x)$. True False

(c) **[2 points]** $f(x) = \begin{cases} \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$ is integrable on $[1,2]$. True False

(d) **[2 points]** $f(x) = \frac{x^2+2x+1}{x+1}$ has removable discontinuity at $x=-1$. True False

(e) **[2 points]** $g(x) = x^4+2x^2-3$ has a minimum value on $[-25,-17]$. True False

NAME: HOPKINS ID:

Quiz 2, Calculus on Manifolds, Spring 2025

JHU Ethics Policy reminder: Each member of the University bears a personal responsibility to uphold the ethical standards of the institution. You should not discuss the content of this quiz with other students in the course until everyone has taken it. Please **sign below** to indicate that you understand the ethics implications in this quiz environment:

You have 60 minutes to answer the following 5 questions:

1. **[5 points]** Define the tangent space at a point on a manifold and state its dimension.

   Solution: Let $x \in M$ be a point in a manifold. The **tangent space** to M at x, denoted T_xM, is the collection of all **tangent vectors**, which we define as linear functionals $D : C^\infty(M) \to \mathbb{R}$ with the property that
   $$D(fg) = g(x)D(f) + f(x)D(g)$$
   for each $f, g \in C^\infty(M)$. The dimension of the tangent space is the same as the manifold.

2. **[10 points]** If $M \subset N$ is a submanifold and $x \in M$, show that $T_xM \subset T_xN$.

   Solution: Every tangent vector arises as the velocity of some curve (as shown in homework 2), thus as every curve in M is also a curve in N the tangent vectors to M at a point x must also be tangent vectors to N at x. Alternatively one can use the fact that the inclusion map $\iota : M \to N$ is smooth and diffeomorphic onto its image (as it is an embedding of M into N), hence its differential $d_x\iota : T_xM \to T_xN$ is injective and provides an inclusion of vector spaces.

3. **[10 points]** Is the cross, $(\mathbb{R} \times \{0\}) \cup (\{0\} \times \mathbb{R}) \subset \mathbb{R}^2$, a manifold? Justify your answer.

   Solution: Notice that away from the origin in $\mathbb{R}^2$ the pieces of the cross are a 1-manifold (as it consists of four half rays diffeomorphic to the open interval $(0,\infty)$); thus if the cross were a manifold it would be 1-dimensional. On the other hand, consider the curves $(t,0)$ and $(0,t)$ for each $t \in \mathbb{R}$. These produce tangent vectors, $(1,0)$ and $(0,1)$, at the origin that span $\mathbb{R}^2$; thus the tangent space at the origin, if it existed, would be 2-dimensional. This contradicts the fact that the tangent space should be the same as that of the manifold (which should be 1-dimensional).

4. Let $\pi : S^n \subset \mathbb{R}^{n+1} \to \mathbb{R}$ be the projection, $\pi(x_1,\dots,x_{n+1}) = x_{n+1}$.

   (i) **[10 points]** Find, by means of the regular value theorem, the values of $y \in \mathbb{R}$ for which $\pi^{-1}(y)$ is an $(n-1)$-manifold. What are the other non-empty $\pi^{-1}(y)$?

   (ii) **[10 points]** For each $y \in \mathbb{R}$ such that $\pi^{-1}(y)$ is an $(n-1)$-manifold, compute the tangent space, $T_x\pi^{-1}(y)$, at each $x \in \pi^{-1}(y)$.

   (iii) **[5 points]** Geometrically, what do each of the non-empty $\pi^{-1}(y)$ correspond to? (hint: draw a picture when $n=2$). Guess which manifolds they are diffeomorphic to.

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NAME: HOPKINS ID:

Quiz 2, Honors Analysis I, Fall 2025

JHU Ethics Policy reminder: Each member of the University bears a personal responsibility to uphold the ethical standards of the institution. You should not discuss the content of this quiz with other students in the course until everyone has taken it. Please **sign below** to indicate that you understand the ethics implications in this quiz environment:

You have 75 minutes to answer the following 6 questions for a total of 100 points:

1. **[20 points]** Let $x_1 > 2$ and define the sequence $(x_n) \subset \mathbb{R}$ by the recursion formula
   $$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right).$$
   Show that the sequence $\{x_n\}_{n\in\mathbb{N}}$ converges, and find its limit.

2. **[15 points]** Let $f : (0,1) \to \mathbb{R}$ be defined by
   $$f(x) = \begin{cases} 0, & \text{if } x \notin \mathbb{Q}, \\ \frac{1}{n}, & \text{if } x = \frac{m}{n}, \text{ where } \gcd(m,n) = 1. \end{cases}$$
   You showed on the homework that f is continuous at every irrational number and discontinuous at each rational in $(0,1)$. Now show however that f is not differentiable at any point of $(0,1)$ (*hint: for $x \in (0,1)$ consider, for each $n \geq 1$, the unique integer a_n such that $nx-1 \leq a_n \leq nx$*).

3. **[20 points]** Suppose that f is differentiable on a closed bounded interval $[a,b]$. Prove that if f' is increasing on (a,b), then f' is continuous on $[a,b]$ (*hint: can a derivative have a jump discontinuity?*).

4. **[20 points]** Suppose that $f_n : E \to \mathbb{R}$ for $n \geq 1$ is a uniformly convergent sequence of functions on a metric space E. If $g : \mathbb{R} \to \mathbb{R}$ is uniformly continuous, prove that the sequence $(g \circ f_n)$ converges uniformly on E.

5. Let ℓ^∞ denote the vector space of all bounded sequences of complex numbers, i.e.
   $$\ell^\infty = \left\{(x_n) \subset \mathbb{C} \;\middle|\; \sup_{n\geq 1}|x_n| < \infty\right\}.$$
   We define a norm on ℓ^∞ by setting
   $$\|(x_n)\|_\infty = \sup_{n\geq 1}|x_n|.$$
   (a) **[5 points]** Prove that $\|\cdot\|_\infty$ is indeed a norm on ℓ^∞.
   (b) **[2 points]** Give an example of a sequence $(x_n) \in \ell^\infty$ with $\|(x_n)\|_\infty = 1$ such that $x_n \neq 1$ for all $n \geq 1$.
   (c) **[3 points]** Is ℓ^∞ a Banach space? Briefly justify.

6. **[15 points]** If you expected a problem to be appear which didn't, state and solve it.

Have a good break!

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NAME: HOPKINS ID:

Quiz 1, PDE I, Fall 2025

JHU Ethics Policy reminder: Each member of the University bears a personal responsibility to uphold the ethical standards of the institution. You should not discuss the content of this quiz with other students in the course until everyone has taken it. Please **sign below** to indicate that you understand the ethics implications in this quiz environment:

You have 60 minutes to answer the following 4 questions for a total of 40 points:

1. (a) **[9 points]** Let $\Omega \subset \mathbb{R}^n$ be open, define what it means for a function $v \in L^1_{\text{loc}}(\Omega)$ to be a weak derivative of $u \in L^1_{\text{loc}}(\Omega)$ corresponding to a multi-index $\alpha \in \mathbb{N}^n$, then define the Sobolev spaces $W^{1,p}(\Omega)$ and $W^{1,p}_0(\Omega)$ for $p \in [1,\infty]$.
   (b) **[6 points]** Find the value of $\lambda \in \mathbb{R}$ so that the function $u \in L^1((-1,1))$ defined by
   $$\begin{cases} u(x) = x+1 & \text{if } x \in (-1,0] \\ u(x) = \lambda - x & \text{if } x \in (0,1) \end{cases},$$
   has a weak derivative $v \in L^1((-1,1))$, and then compute this weak derivative.

2. (a) **[2 points]** Define what it means for a function to be absolutely continuous.
   (b) **[4 points]** Use (a) to state (but not prove) a characterisation of the Sobolev spaces $W^{1,p}((a,b))$ for bounded intervals $(a,b) \subset \mathbb{R}$ and $p \in [1,\infty]$.
   (c) **[2 points]** Can a continuous but nowhere differentiable function on a bounded interval have a weak derivative?

3. (a) **[1 points]** Define the Sobolev conjugate, p^*, of $p \in [1,n)$.
   (b) **[3 points]** State the Gagliardo–Nirenberg–Sobolev inequality for functions in $C^1_c(\mathbb{R}^n)$.
   (c) **[5 points]** Prove that if $u \in C^1_c(\mathbb{R}^2) \cap W^{1,2}(\mathbb{R}^2)$, then for some constant $C > 0$ independent of u we have
   $$\int u^4 \leq C\left(\int u^2\right)\left(\int |Du|^2\right),$$
   where all integrals are over $\mathbb{R}^2$ (*hint: use (b) and consider u^2*).
   (d) **[2 points]** Let $\Omega \subset \mathbb{R}^n$ be open with $\partial\Omega$ C^1 and state the Gagliardo–Nirenberg–Sobolev inequality for functions in $W^{1,p}(\Omega)$.
   (e) **[1 points]** Prove that if $\Omega \subset \mathbb{R}^2$ is open with $\partial\Omega$ C^1 and $u \in W^{1,2}(\Omega)$, then there is a constant $C > 0$ depending on Ω but independent of u such that
   $$\int u^4 \leq C\left(\int u^2\right)\left(\int |Du|^2\right) + C\left(\int u^2\right)^2,$$
   where all integrals are over Ω (*hint: use (d) and consider u^2*).

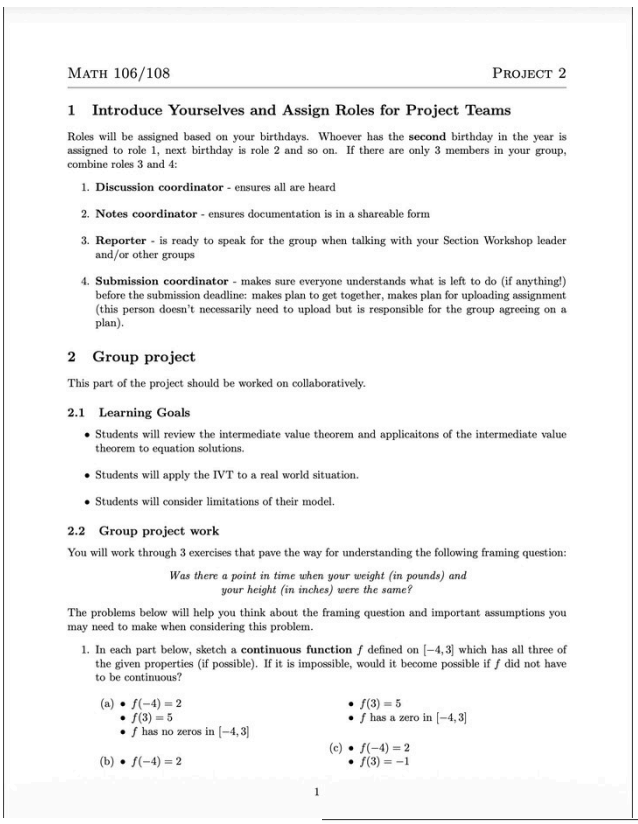
4. **[5 points]** If you expected a problem to be appear which didn't, state and solve it.

Have a good break!

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3.3 Group & individual projects

Screenshots of a group and individual project for Calculus I are included below.



MATH 106/108 PROJECT 2

1 Introduce Yourselves and Assign Roles for Project Teams

Roles will be assigned based on your birthdays. Whoever has the **second** birthday in the year is assigned to role 1, next birthday is role 2 and so on. If there are only 3 members in your group, combine roles 3 and 4:

1. **Discussion coordinator** - ensures all are heard
2. **Notes coordinator** - ensures documentation is in a shareable form
3. **Reporter** - is ready to speak for the group when talking with your Section Workshop leader and/or other groups
4. **Submission coordinator** - makes sure everyone understands what is left to do (if anything!) before the submission deadline: makes plan to get together, makes plan for uploading assignment (this person doesn't necessarily need to upload but is responsible for the group agreeing on a plan).

2 Group project

This part of the project should be worked on collaboratively.

2.1 Learning Goals

- Students will review the intermediate value theorem and applicaitons of the intermediate value theorem to equation solutions.
- Students will apply the IVT to a real world situation.
- Students will consider limitations of their model.

2.2 Group project work

You will work through 3 exercises that pave the way for understanding the following framing question:

Was there a point in time when your weight (in pounds) and your height (in inches) were the same?

The problems below will help you think about the framing question and important assumptions you may need to make when considering this problem.

1. In each part below, sketch a **continuous function** f defined on $[-4, 3]$ which has all three of the given properties (if possible). If it is impossible, would it become possible if f did not have to be continuous?

   (a)
   - $f(-4) = 2$
   - $f(3) = 5$
   - f has no zeros in $[-4, 3]$

   (b)
   - $f(-4) = 2$
   - $f(3) = 5$
   - f has a zero in $[-4, 3]$

   (c)
   - $f(-4) = 2$
   - $f(3) = -1$
   - f has no zeros in $[-4, 3]$

   (d)
   - $f(-4) = 2$
   - $f(3) = -1$
   - f does not attain the value 1 on $[-4, 3]$

2. Let $f(x) = \sin(x) + x + 2$ and $g(x) = \log(3x + 1) + x^3 - 3$. Is there a value of x between 0 and 5 for which $f(x) = g(x)$?

3. What is the average height and weight of a new born? What is the average height and weight of an adult? Make sure to cite your references!

2.3 Group Submission - Due Friday 9/27/2024 by 11:59pm EST on Gradescope

In a concise one page report, answer the framing question:

Was there a point in time when your weight (in pounds) and your height (in inches) were the same?

and point to your work in 1, 2, and 3 to support your claim and to explain why you need to make the key assumption you had to make. Your submission should include a discussion of what assumption you must make about the height and weight functions to apply the IVT and whether or not your group thinks that this is a reasonable assumption to make. Make sure to properly cite (using a citation style of your choice) all references you use.

To submit this assignment, you will need a scanned copy of your project saved as a PDF via **Gradescope**. If you write out your assignment on paper, you can create a PDF file by using a scanning app (e.g. the Gradescope app) on your phone or tablet, or using a scanner in the library. Once you have your PDF copy of your homework, head to the course Canvas page and select **Gradescope** from the left hand menu. Once in **Gradescope** select the correct assignment to turn in (e.g. "Group Project 2") and follow the prompts to upload your PDF. It will ask you to **tag the problems on each page of your submission** so the graders can find your work (don't skip this step!). After you have tagged the PDF you are ready to submit.

3 Individual project

Sometimes odd behavior can be hidden behind the surface of rather normal-looking functions. Consider the following function:

$$f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 0 & \text{if } x \geq 0 \end{cases} \tag{1}$$

Sketch a graph of this function. Does it have any vertical segments or corners? Is it differentiable everywhere? If so, sketch the derivative of this function. Is the derivative differentiable everywhere? If not, at what point(s) is it not differentiable? Draw the second derivative wherever it exists. Is the second derivative differentiable? Is it continuous?

3.1 Individual writing assignment for portfolio 1

****106**** You are hired by Pearson to update and edit a 5th edition of Calculus for Biology and Medicine by Claudia Neuhauser and Marcus L. Roper. You decide that the problem above would make for a good example following example 1 in section 4.2.2 of the text before the introduction of the Theorem. Rewrite this problem and complete solution(s) with explanation for inclusion in edition 5 of the text. Your audience for this assignment is your group of peers in the class. Your example and solution should not exceed one page.

Notice that in example 1, the textbook does not clearly state what question is being answered by the example. It would have been better for there to be an explicit question at the start, asked in a simple way such as

- Where is $g(x) = |x|$ differentiable?

Your write-up should state the question you are answering in a simple way in addition to providing a solution. (Example 3 in Section 4.1 does this well.)

Notice that the solution gives two perspectives:

- algebraic (computing using the limit definition of the derivative) and,
- geometric (looking at the graphs of and its derivative!)

Your solution should be similarly scaffolded for your peers to use as a learning tool and be able to be inserted into the section after example 1.

****108**** You are hired by Cengage Learning to update and edit a 9th edition of Single Variable Calculus, Early Transcendentals by James Stewart. You decide that the problem above would make for a good example following example 5 in section 2.8 of the text before the introduction of Theorem 4. Rewrite this problem and complete solution(s) with explanation for inclusion in edition 9 of the text. Your audience for this assignment is your group of peers in the class. Your example and solution should not exceed one page.

Notice that in an Example like 5, the question is asked in a simple way:

- Where is $g(x) = |x|$ differentiable?

But the solution gives two perspectives:

- algebraic (computing using the limit definition of the derivative) and,
- geometric (looking at the graphs of and its derivative!)

Your solution should be similarly scaffolded for your peers to use as a learning tool and be able to be inserted into the section after example 5.

3.4 Presentations

For the (graduate level) Partial Differential Equations II course, instead of written assignments, I asked students to give a 25 minute presentation on a topic of their choosing. The suggested topic list, including those that were needed to round out course content as well as optional extras for graduate students, was:

Needed:

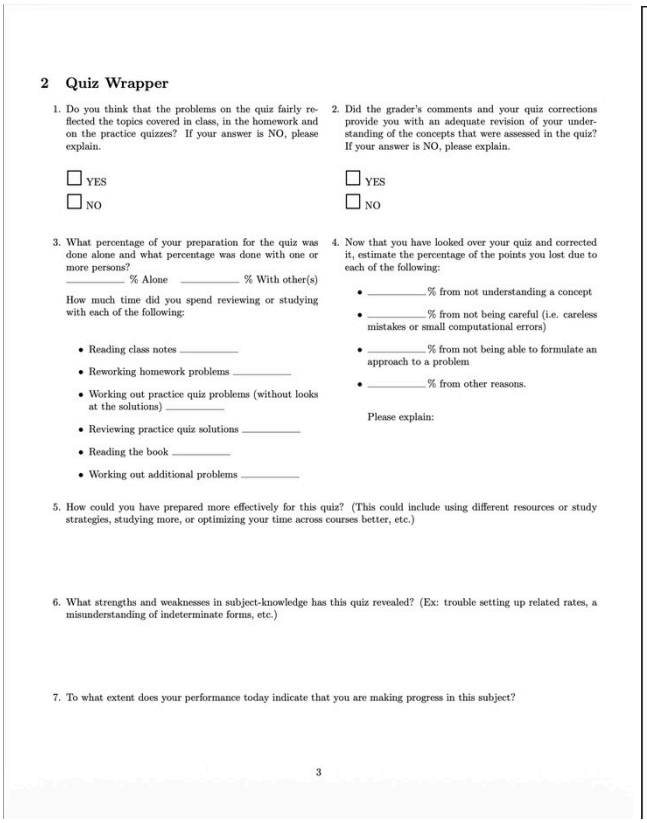
- Calderón--Zygmund estimates for the Poisson equation .
- Construction of barriers for the MSE .
- Non-existence for the MSE on non-convex domains .

Optional:

- Parabolic PDE (i.e. how does the theory of PDE I+II adapt to this setting).
- Mountain pass theorem and applications to semilinear elliptic PDE.
- Curvature estimates for minimal hypersurfaces (by Di Giorgi iteration).
- Moser iteration (as an alternative approach to Di Giorgi iteration).
- Smooth to analytic for solutions to elliptic PDE.

3.5 Reflections

Screenshots of quiz wrappers and course reflections for Calculus I are included below.



2 Quiz Wrapper

1. Do you think that the problems on the quiz fairly reflected the topics covered in class, in the homework and on the practice quizzes? If your answer is NO, please explain.

☐ YES
☐ NO

2. Did the grader's comments and your quiz corrections provide you with an adequate revision of your understanding of the concepts that were assessed in the quiz? If your answer is NO, please explain.

☐ YES
☐ NO

3. What percentage of your preparation for the quiz was done alone and what percentage was done with one or more persons?
_____ % Alone _____ % With other(s)

How much time did you spend reviewing or studying with each of the following:

- Reading class notes _____
- Reworking homework problems _____
- Working out practice quiz problems (without looks at the solutions) _____
- Reviewing practice quiz solutions _____
- Reading the book _____
- Working out additional problems _____

4. Now that you have looked over your quiz and corrected it, estimate the percentage of the points you lost due to each of the following:

- _____% from not understanding a concept
- _____% from not being careful (i.e. careless mistakes or small computational errors)
- _____% from not being able to formulate an approach to a problem
- _____% from other reasons.

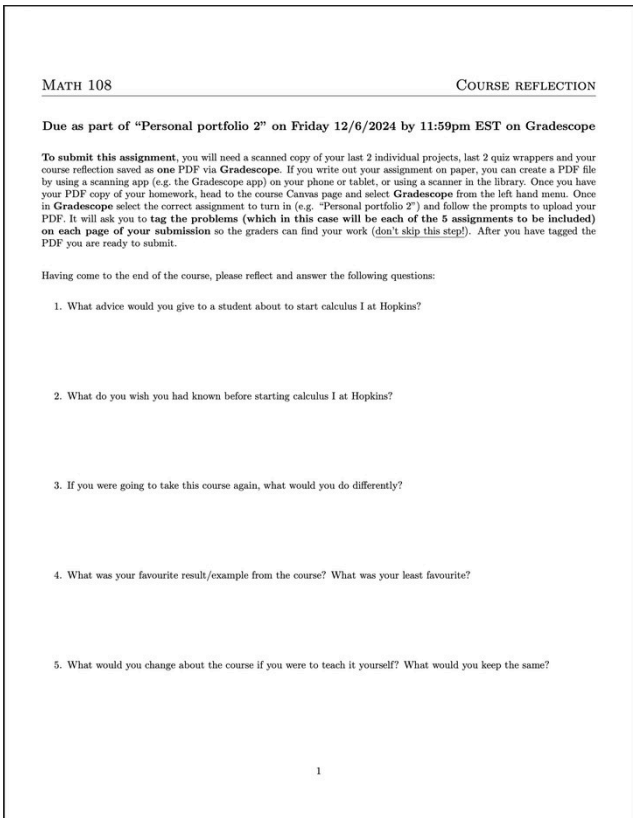
Please explain:

5. How could you have prepared more effectively for this quiz? (This could include using different resources or study strategies, studying more, or optimizing your time across courses better, etc.)

6. What strengths and weaknesses in subject-knowledge has this quiz revealed? (Ex: trouble setting up related rates, a misunderstanding of indeterminate forms, etc.)

7. To what extent does your performance today indicate that you are making progress in this subject?

3



MATH 108 COURSE REFLECTION

Due as part of "Personal portfolio 2" on Friday 12/6/2024 by 11:59pm EST on Gradescope

To submit this assignment, you will need a scanned copy of your last 2 individual projects, last 2 quiz wrappers and your course reflection saved as **one** PDF via **Gradescope**. If you write out your assignment on paper, you can create a PDF file by using a scanning app (e.g. the Gradescope app) on your phone or tablet, or using a scanner in the library. Once you have your PDF copy of your homework, head to the course Canvas page and select **Gradescope** from the left hand menu. Once in **Gradescope** select the correct assignment to turn in (e.g. "Personal portfolio 2") and follow the prompts to upload your PDF. It will ask you to **tag the problems (which in this case will be each of the 5 assignments to be included) on each page of your submission** so the graders can find your work (don't skip this step!). After you have tagged the PDF you are ready to submit.

Having come to the end of the course, please reflect and answer the following questions:

1. What advice would you give to a student about to start calculus I at Hopkins?

2. What do you wish you had known before starting calculus I at Hopkins?

3. If you were going to take this course again, what would you do differently?

4. What was your favourite result/example from the course? What was your least favourite?

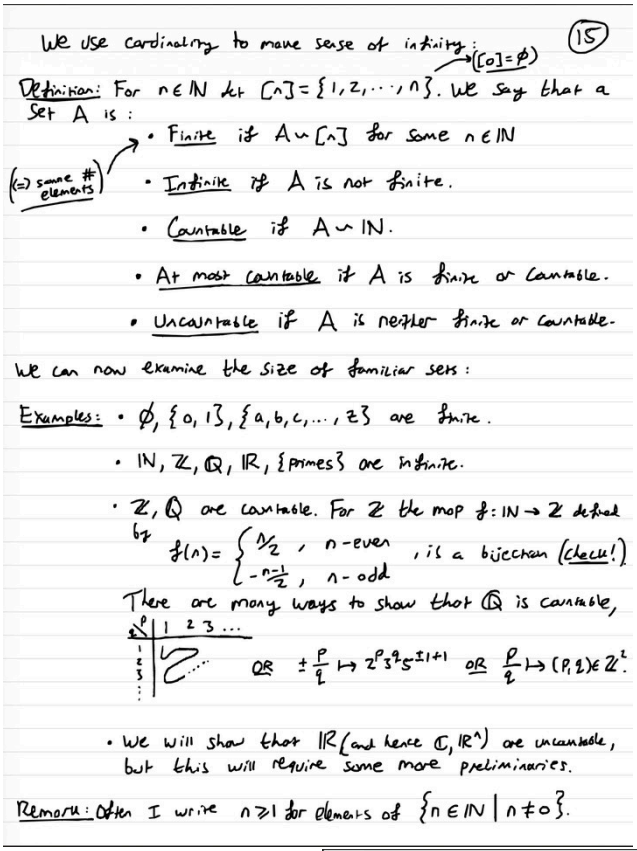
5. What would you change about the course if you were to teach it yourself? What would you keep the same?

1

4 Note samples

4.1 Handwritten

Students were provided with handwritten notes for the three courses Honors Analysis I, as well as Partial Differential Equations I + II; a screenshot from each is included below.



We use cardinality to make sense of infinity: (15)

Definition: For $n \in \mathbb{N}$ let $[n] = \{1, 2, \cdots, n\}$. ($[0] = \emptyset$) We say that a set A is:

- **Finite** if $A \sim [n]$ for some $n \in \mathbb{N}$ ($\Leftrightarrow$ same # elements)
- **Infinite** if A is not finite.
- **Countable** if $A \sim \mathbb{N}$.
- **At most countable** if A is finite or countable.
- **Uncountable** if A is neither finite or countable.

We can now examine the size of familiar sets:

Examples:
- $\emptyset, \{0, 1\}, \{a, b, c, \ldots, z\}$ are finite.
- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \{\text{primes}\}$ are infinite.
- $\mathbb{Z}, \mathbb{Q}$ are countable. For $\mathbb{Z}$ the map $f: \mathbb{N} \to \mathbb{Z}$ defined by $f(n) = \begin{cases} n/2, & n \text{ even} \\ -\frac{n-1}{2}, & n \text{ odd} \end{cases}$, is a bijection (check!)

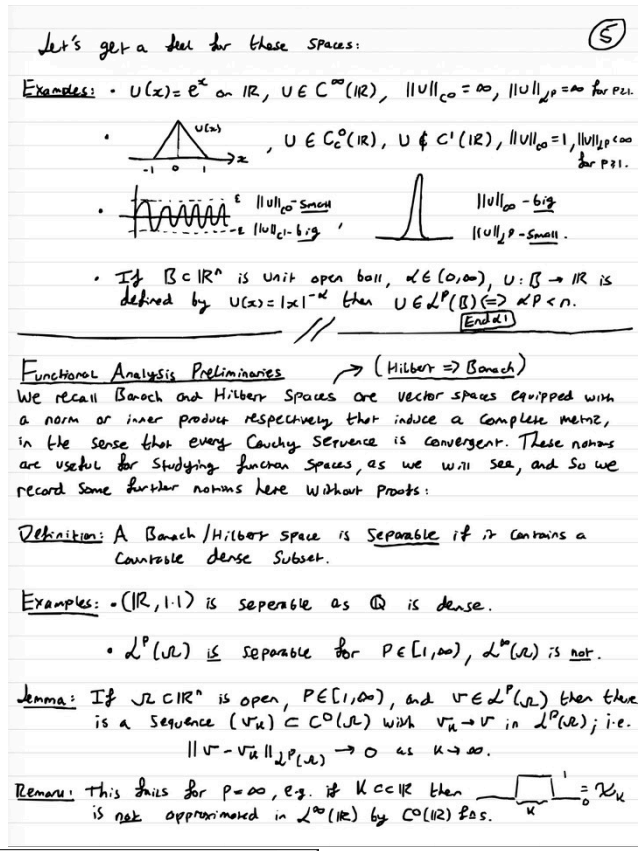
  There are many ways to show that $\mathbb{Q}$ is countable,

p \ q	1	2	3	…
1				
2				
3				
⋮				

  OR $\pm \frac{p}{q} \mapsto 2^p 3^q 5^{|1|+1}$ OR $\frac{p}{q} \mapsto (p, q) \in \mathbb{Z}^2$.

- We will show that $\mathbb{R}$ (and hence $\mathbb{C}, \mathbb{R}^n$) are uncountable, but this will require some more preliminaries.

Remark: Often I write $n \geq 1$ for elements of $\{n \in \mathbb{N} \mid n \neq 0\}$.



Let's get a feel for these spaces: (5)

Examples:
- $u(x) = e^x$ on $\mathbb{R}$, $u \in C^\infty(\mathbb{R})$, $\|u\|_{C^0} = \infty$, $\|u\|_{L^p} = \infty$ for $p \geq 1$.
-

  , $u \in C_c^0(\mathbb{R})$, $u \notin C^1(\mathbb{R})$, $\|u\|_{C^0} = 1$, $\|u\|_{L^p} < \infty$ for $p \geq 1$.
-

  $\|u\|_{C^0}$ - small, $\|u\|_{C^1}$ - big, $\|u\|_\infty$ - big, $\|u\|_{L^p}$ - small.
- If $B \subset \mathbb{R}^n$ is unit open ball, $\alpha \in (0, \infty)$, $U: B \to \mathbb{R}$ is defined by $U(x) = |x|^{-\alpha}$ then $U \in L^p(B) \Leftrightarrow \alpha p < n$. End of 1

Functional Analysis Preliminaries ($\to$ Hilbert $\Rightarrow$ Banach)

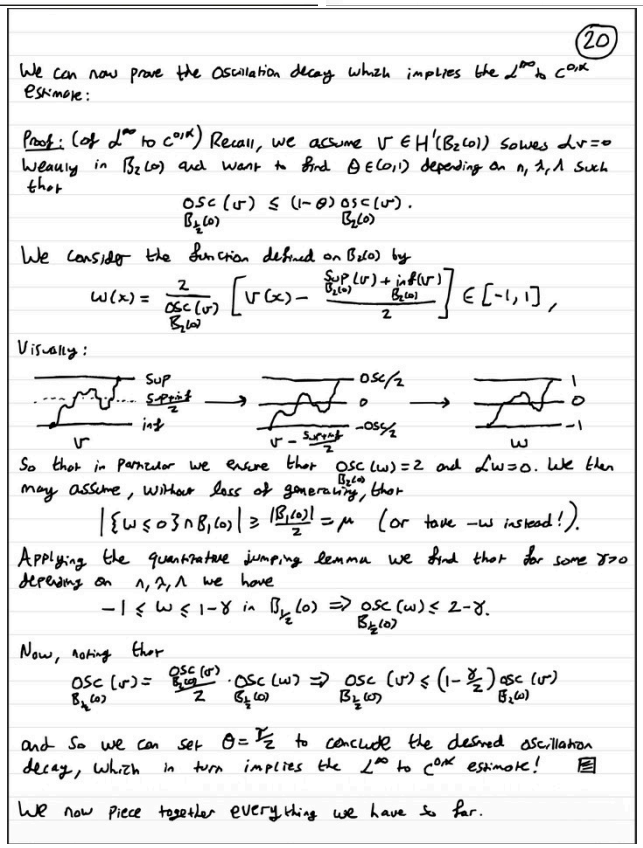
We recall Banach and Hilbert spaces are vector spaces equipped with a norm or inner product respectively that induce a complete metric, in the sense that every Cauchy sequence is convergent. These norms are useful for studying function spaces, as we will see, and so we record some further notions here without proofs:

Definition: A Banach/Hilbert space is **separable** if it contains a countable dense subset.

Examples:
- $(\mathbb{R}, |\cdot|)$ is separable as $\mathbb{Q}$ is dense.
- $L^p(\mathbb{R})$ is separable for $p \in [1, \infty)$, $L^\infty(\mathbb{R})$ is not.

Lemma: If $\Omega \subset \mathbb{R}^n$ is open, $p \in [1, \infty)$, and $v \in L^p(\Omega)$ then there is a sequence $(v_k) \subset C^0(\Omega)$ with $v_k \to v$ in $L^p(\Omega)$; i.e. $\|v - v_k\|_{L^p(\Omega)} \to 0$ as $k \to \infty$.

Remark: This fails for $p = \infty$, e.g. if $K \subset\subset \mathbb{R}$ then is not approximated in $L^\infty(\mathbb{R})$ by $C^0(\mathbb{R})$ fcts.



(20)

We can now prove the oscillation decay which implies the L^∞ to $C^{0,\alpha}$ estimate:

Proof: (of L^∞ to $C^{0,\alpha}$) Recall, we assume $v \in H^1(B_2(0))$ solves $\Delta v = 0$ weakly in $B_2(0)$ and want to find $\theta \in (0,1)$ depending on n, λ, Λ such that

$$\operatorname{osc}_{B_{\frac{1}{2}}(0)} (v) \leq (1 - \theta) \operatorname{osc}_{B_2(0)} (v).$$

We consider the function defined on $B_2(0)$ by

$$w(x) = \frac{2}{\operatorname{osc}_{B_2(0)} (v)} \left[v(x) - \frac{\sup_{B_2(0)} (v) + \inf_{B_2(0)} (v)}{2} \right] \in [-1, 1],$$

Visually:

So that in particular we ensure that $\operatorname{osc}_{B_2(0)} (w) = 2$ and $\Delta w = 0$. We then may assume, without loss of generality, that

$$|\{w \leq 0\} \cap B_1(0)| \geq \frac{|B_1(0)|}{2} = \mu \quad \text{(or take } -w \text{ instead!)}.$$

Applying the quantitative jumping lemma we find that for some $\gamma > 0$ depending on n, λ, Λ we have

$$-1 \leq w \leq 1 - \gamma \text{ in } B_{\frac{1}{2}}(0) \Rightarrow \operatorname{osc}_{B_{\frac{1}{2}}(0)} (w) \leq 2 - \gamma.$$

Now, noting that

$$\operatorname{osc}_{B_{\frac{1}{2}}(0)} (v) = \frac{\operatorname{osc}_{B_2(0)} (v)}{2} \cdot \operatorname{osc}_{B_{\frac{1}{2}}(0)} (w) \Rightarrow \operatorname{osc}_{B_{\frac{1}{2}}(0)} (v) \leq \left(1 - \frac{\gamma}{2}\right) \operatorname{osc}_{B_2(0)} (v)$$

and so we can set $\theta = \frac{\gamma}{2}$ to conclude the desired oscillation decay, which in turn implies the L^∞ to $C^{0,\alpha}$ estimate! ∎

We now piece together everything we have so far.

4.2 Typed

For the Calculus on Manifolds course I typed up notes myself; a screenshot of two pages is included below:



1.2 What is (and is not) a manifold?

The concepts of calculus make sense locally, by which we mean one only needs to know how a function behaves near a given point. We can thus reasonably expect to make sense of those same concepts in more general spaces where one only asks for a Euclidean structure locally, allowing for the global picture to potentially look very different from Euclidean space. A slightly more precise formulation of what it means to be locally Euclidean is given by the notion of coordinates; for example, near any point on the sphere one needs only two numbers to smoothly parametrise the space.

Loosely, a manifold will be a space formed by patching together small regions in which the object has a Euclidean structure. This is in strong analogy with the way in which maps of the earth are created through the use of charts and atlases (terms we will soon see in a mathematical context). Before diving into the formal definition of a manifold, we first familiarise ourselves with some examples (and non-examples) of manifolds with the idea in mind that a manifold should be an object composed of regions that can be smoothly deformed to look like Euclidean space.

1.2.1 Examples of manifolds in Euclidean space

We first look at some examples of manifolds that arise as subsets of Euclidean space:

- For each non-negative integer, n, the **n-dimensional Euclidean space**, $\mathbb{R}^n$, is an n-dimensional manifold, hereafter an n-manifold, as is any open subset, $U \subset \mathbb{R}^n$. If $n = 0$ then $\mathbb{R}^0$ is a point, unions of points are 0-manifolds.
- Any m-dimensional subspace (or affine plane) of $\mathbb{R}^n$ is an m-manifold. Necessarily $m \leq n$, $m = 1$ corresponds to lines, and $m = 2$ corresponds to planes.
- The circle, $S^1 = \{(x_1, x_2) \in \mathbb{R}^2 \,|\, x_1^2 + x_2^2 = 1\} \subset \mathbb{R}^2$, is a 1-manifold (explicit coordinates are given by $\theta \mapsto (\cos(\theta), \sin(\theta))$ and the sphere, $S^2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \,|\, x_1^2 + x_2^2 + x_3^2 = 1\} \subset \mathbb{R}^3$, is a 2-manifold (coordinates can be given by stereographic projection). More generally, the n-sphere, $S^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \,|\, \sum_{i=1}^n x_i^2 = 1\} \subset \mathbb{R}^{n+1}$, is an n-manifold.
- The (donut) torus, given by $\{((2 + \cos(\theta))\cos(\phi), (2 + \cos(\theta))\sin(\phi), \sin(\theta)) \,|\, \theta, \phi \in \mathbb{R}\} \subset \mathbb{R}^2$, is a 2-manifold. The (standard) torus, $T^2 = S^1 \times S^1 \subset \mathbb{R}^4$, is also a 2-manifold. These two object, while different as sets, are the 'same' manifold in an appropriate sense (they are diffeomorphic).
- The Möbius band, $B = \{(\cos(\theta), \sin(\theta), r\cos(\frac{\theta}{2}), r\sin(\frac{\theta}{2}) \in \mathbb{R}^4 \,|\, r, \theta \in \mathbb{R}\} \subset \mathbb{R}^4$, is a 2-manifold. Notice that projection to the first two coordinates gives a copy of the circle, S^1, whose preimage is a straight line. As θ varies over an interval of length 2π this line rotates a half turn (so two loops round the circle in the projection arrive at the same point).

1.2.2 Examples of abstract manifolds

Not all manifolds sit inside of Euclidean space in a natural way. For example, in physics one thinks of our universe as a 4-manifold (three dimensions of space and one of time), but we do not seem to exist as a subset of some bigger ambient space. We now look at examples of abstract manifolds (abstract in the sense that they do not lie naturally in a copy of $\mathbb{R}^n$):

- Space-time itself as a 4-manifold does not seem to sit inside of some copy of $\mathbb{R}^n$, but is locally Euclidean.

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- Consider the matrix groups, $GL(n, \mathbb{R})$ and $SL(n, \mathbb{R})$, formed of $n \times n$ matrices with non-zero and unit determinant respectively. These form manifolds of dimension n^2 and $n^2 - 1$ respectively. While these groups can be viewed as subsets of $\mathbb{R}^{n^2}$ (for instance by stacking columns of the matrices), this perspective does not tell us much, if anything, about their group structure.
- The (Pac-Man) torus, formed by identifying opposite sides of a square (with the same orientation) is a 2-manifold. This does not sit inside of $\mathbb{R}^3$, like the donut torus, without having to 'bend/twist' the square.
- The Klein bottle, formed by identifying two sides of a square with the same orientation and identifying the other two sides with opposite orientation, is a 2-manifold. There is no way to make this sit in $\mathbb{R}^3$ without it self-intersecting.
- The projective space, $\mathbb{R}P^n$, is formed by either taking the n-sphere, S^n, and identifying antipodal points or by identifying colinear points in $\mathbb{R}^{n+1}$. This is an n-manifold and may be thought of as the space of lines in $\mathbb{R}^{n+1}$.

The final three examples of abstract manifolds above are formed as quotient spaces of Euclidean space. As it turns out, and as we will see later on, every manifold can realised as a subset of $\mathbb{R}^n$ but this is not always a natural way to view them, and may not preserve any additional structres on the manfiold (such as a notion of distance). We will thus be interested in definitions and concepts that are intrinsic to the manifold itself, and do not make reference to an ambient space.

1.2.3 Examples that are not manifolds

Let us finally discuss some examples of objects that we want to preclude as manifolds:

- Consider a figure of eight (or just the union of the coordinate axes in $\mathbb{R}^2$). While this is a 1-manifold away from the intersection point, at this intersection point there is no way to assign local coordinates there. Similarly, the cone, $C = \{(x_1, x_2, x_3) \,|\, x_1^2 + x_2^2 - x_3^2 = 0\}$ is not a manifold.
- Consider the square or cube. while the faces of the shapes locally look like $\mathbb{R}^2$, the corners of the object are not smooth. These are examples of topological but not smooth manifolds (which we will refer to simply as manifolds in this course).
- Fractal like objects such as the Koch snowflake, Peano space filling curves, and Alexander's horned sphere (homeomorphic but not diffeomorphic to the sphere) are locally homeomorphic to Euclidean space but not smooth in an appropriate sense.

The second and third bullet points above describe objects that are topologically manifolds in the sense that they locally can be deformed to look like Euclidean space, but this deformation cannot be done in a smooth manner; and hence they are not (smooth) manifolds.

2 Manifold preliminaries

We want to identify an appropriate mathematical notion that allows us to piece together sets that look like Euclidean space into a global object. This will be done by first introducing topological spaces, sometimes referred to 'rubber sheet geometry', which posses the desired property of the abstract

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Proposition 5. *(Properties of the global differential) Let $F : M \to N$ and $G : N \to P$ be smooth maps between manifolds. Then we have the following:*

1. *$dF : TM \to TN$ is smooth as a map between manifolds.*
2. *$d(G \circ F) = dG \circ dF$.*
3. *$d(I_M) = I_{TM}$.*
4. *If F is a diffeomorphism then dF is a diffeomorphism with $(dF)^{-1} = d(F^{-1})$.*

Proof. The first property follows by virtue of the fact that the coordinate representation of d_xF depends smoothly on the point $x \in M$. The second, third, and fourth property follow directly from the properties of the differential previously established. □

Just as we alternatively called the differential of a smooth map at a point a pushforward, we have the following notion:

Definition 31. *If $F : M \to N$ is a diffeomorphism between manifolds, then the **pushforward** of F, denoted $F_* : \Gamma(TM) \to \Gamma(TN)$, is defined for $X \in \Gamma(TM)$ by $F_*(X) = dF(X)$; i.e. $F_*(X)(F(x)) = d_xF(X_x)$ for each $x \in M$.*

Remark 33. *We require F to be a diffeomorphism in the definition above since if F were not injective then it would not be well defined, and if F was not surjective then it would not be defined on all of N.*

3.5 Cotangent bundle, 1-forms, and the line integral

A physical interpretation of the integral along a curve in Euclidean space is that of the 'work' done moving along the curve. Precisely, given a smooth curve $\gamma : [a, b] \to \mathbb{R}^n$ with $\gamma(a) = x$ and $\gamma(b) = y$, the integral along this curve should provide us with a number quantifying how much 'work' (or energy etc.) was done moving from x to y along γ. We can discretise this curve by considering vectors, v_i, joining points, x_{i-1} to x_i, which lie on γ so that $x_0 = a$ and $x_n = b$, and then assign an amount of 'work', $\omega_i \in \mathbb{R}$, taken to move along the vector v_i from x_{i-1} to x_i. The 'work' done along this discrete path of vectors from a to b is then equal to $\sum_{i=1}^n \omega_i$. If we imagine taking a limit of this process we would need an object, ω, which inputs vectors tangent to the curve and outputs a number at each point of γ. We should then have by a limiting process that the 'work' done to move from a to b along γ, or the integral of ω along γ, is given by

$$\int_\gamma \omega = \lim_{n \to \infty} \sum_{i=1}^n \omega_i.$$

See [Tao20] for more motivation in this vein (which is where I took the idea for this brief motivation from). In this section we will make this above idea precise.

In order to make sense of the above discussion we will introduce objects that are "dual" to tangent vectors and fields, for which we need to recall some linear algebra:

Definition 32. *Given a finite dimensional vector space, V, over $\mathbb{R}$, a **covector** is a linear functional $\omega : V \to \mathbb{R}$; i.e. a linear map from V to $\mathbb{R}$. The space of all covectors is the **dual space** of V, which we denote by V^*. Given a basis, $\{e_i\}_{i=1}^n \subset V$, of V we define the **dual basis**, $\{\mathcal{E}^i\}_{i=1}^n \subset V^*$, for V^**

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A full version of the notes, which I will update for the Spring 2027 course, are maintained at:

`https://kobe-marshall-stevens.github.io/Calculus_on_Manifolds_notes.pdf`

5 Evaluations & Feedback samples

In this section I include a sample of some of the evaluation statistics, specific feedback, and also anonymous responses to a form I send to each class. At Johns Hopkins, for each of my (non graduate level) courses, students are made to submit evaluations and feedback about the course before they can view their final grades. The full list of evaluations & feedback for these courses is available upon request.

5.1 Course evaluations

For the Calculus I course (there were 4 sections with 21-31 students in each):

1 - The overall quality of this course is:

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	0	0.00%
Satisfactory	(3)	3	9.68%
Good	(4)	13	41.94%
Excellent	(5)	15	48.39%
N/A	(0)	0	0.00%

Percent Responses: 0 25 50 100

Means: Question 4.39 | School Level 4.16 | Department Level 3.63

Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
31/31 (100.00%)	4.39	0.67	4.00	18800	4.16	0.95	4.00	1400	3.63	1.21	4.00

2 - The instructor's teaching effectiveness is:

Kobe Marshall-Stevens

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	2	6.45%
Satisfactory	(3)	4	12.90%
Good	(4)	7	22.58%
Excellent	(5)	18	58.06%
N/A	(0)	0	0.00%

Percent Responses: 0 25 50 100

Means: Question 4.32 | School Level 4.16 | Department Level 3.57

Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
31/31 (100.00%)	4.32	0.94	5.00	22429	4.16	1.03	4.00	1383	3.57	1.33	4.00

1 - The overall quality of this course is:

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	0	0.00%
Satisfactory	(3)	3	15.00%
Good	(4)	7	35.00%
Excellent	(5)	10	50.00%
N/A	(0)	0	0.00%

Percent Responses: 0 25 50 100

Means: Question 4.35 | School Level 4.16 | Department Level 3.63

Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
20/21 (95.24%)	4.35	0.75	4.50	18800	4.16	0.95	4.00	1400	3.63	1.21	4.00

2 - The instructor's teaching effectiveness is:

Kobe Marshall-Stevens

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	1	5.26%
Satisfactory	(3)	0	0.00%
Good	(4)	6	31.58%
Excellent	(5)	12	63.16%
N/A	(0)	0	0.00%

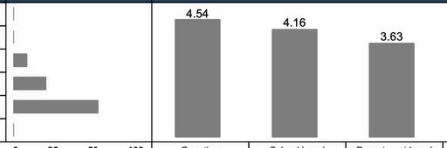
Percent Responses: 0 25 50 100

Means: Question 4.53 | School Level 4.16 | Department Level 3.57

Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
19/21 (90.48%)	4.53	0.77	5.00	22429	4.16	1.03	4.00	1383	3.57	1.33	4.00

1 - The overall quality of this course is:

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	0	0.00%
Satisfactory	(3)	3	10.71%
Good	(4)	7	25.00%
Excellent	(5)	18	64.29%
N/A	(0)	0	0.00%

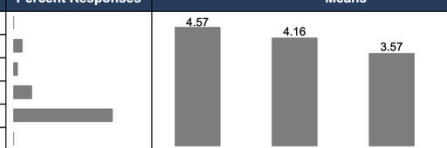


Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
28/28 (100.00%)	4.54	0.69	5.00	18800	4.16	0.95	4.00	1400	3.63	1.21	4.00

2 - The instructor's teaching effectiveness is:

Kobe Marshall-Stevens

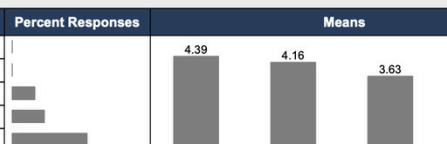
Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	2	7.14%
Satisfactory	(3)	1	3.57%
Good	(4)	4	14.29%
Excellent	(5)	21	75.00%
N/A	(0)	0	0.00%



Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
28/28 (100.00%)	4.57	0.88	5.00	22429	4.16	1.03	4.00	1383	3.57	1.33	4.00

1 - The overall quality of this course is:

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	0	0.00%
Satisfactory	(3)	5	17.86%
Good	(4)	7	25.00%
Excellent	(5)	16	57.14%
N/A	(0)	0	0.00%



Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
28/28 (100.00%)	4.39	0.79	5.00	18800	4.16	0.95	4.00	1400	3.63	1.21	4.00

2 - The instructor's teaching effectiveness is:

Kobe Marshall-Stevens

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	0	0.00%
Satisfactory	(3)	3	11.54%
Good	(4)	8	30.77%
Excellent	(5)	15	57.69%
N/A	(0)	0	0.00%

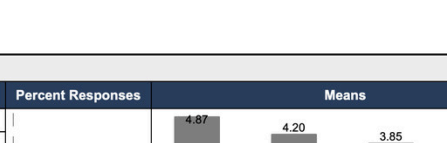


Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
26/28 (92.86%)	4.46	0.71	5.00	22429	4.16	1.03	4.00	1383	3.57	1.33	4.00

For the Calculus on Manifolds course:

1 - The overall quality of this course is:

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	0	0.00%
Satisfactory	(3)	0	0.00%
Good	(4)	2	13.33%
Excellent	(5)	13	86.67%
N/A	(0)	0	0.00%

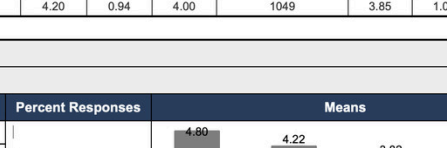


Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
15/15 (100.00%)	4.87	0.35	5.00	17229	4.20	0.94	4.00	1049	3.85	1.05	4.00

2 - The instructor's teaching effectiveness is:

Kobe Marshall-Stevens

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	0	0.00%
Satisfactory	(3)	1	6.67%
Good	(4)	1	6.67%
Excellent	(5)	13	86.67%
N/A	(0)	0	0.00%



Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
15/15 (100.00%)	4.80	0.56	5.00	20606	4.22	1.00	5.00	1041	3.82	1.15	4.00

For the Honors Analysis I course:

1 - The overall quality of this course is:

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	0	0.00%
Satisfactory	(3)	1	5.26%
Good	(4)	2	10.53%
Excellent	(5)	16	84.21%
N/A	(0)	0	0.00%

Percent Responses: 0 25 50 100

Means: 4.79 (Question), 4.20 (School Level), 3.82 (Department Level)

Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
19/20 (95.00%)	4.79	0.54	5.00	18448	4.20	0.93	4.00	1329	3.82	1.02	4.00

2 - The instructor's teaching effectiveness is:

Kobe Marshall-Stevens

Response Option	Weight	Frequency	Percent
Poor	(1)	0	0.00%
Weak	(2)	1	5.26%
Satisfactory	(3)	0	0.00%
Good	(4)	2	10.53%
Excellent	(5)	16	84.21%
N/A	(0)	0	0.00%

Percent Responses: 0 25 50 100

Means: 4.74 (Question), 4.18 (School Level), 3.67 (Department Level)

Response Rate	Mean	STD	Median	School Level	Mean	STD	Median	Department Level	Mean	STD	Median
19/20 (95.00%)	4.74	0.73	5.00	22517	4.18	1.01	4.00	1326	3.67	1.18	4.00

5.2 Course feedback

For the Calculus on Manifolds course:

8 - What are the best aspects of this course?

Response Rate	11/15 (73.33%)

- The lectures are very enjoyable, Kobe does a great job explaining things. I also really like the topic.
- The professor is phenomenal. By far the best math teacher I have had, and the time and commitment he gives to our learning really shows. The content is very difficult but the Professor makes a great effort to make sure we understand it.
- The professor is very nice and also tends to give good grades.
- Very well organised course, the professor is very nice, explains everything clearly, very interesting course
- The professor is a good lecturer and also very kind.
- Kobe is a very good lecturer. The course is very organized and he made the content sufficiently difficult so that it would not be boring for those who already had some background in geometry. The lecture notes are also fantastic and the grading criteria and quizess are fair.
- The professor is so understanding and really cares that you get the material. Office hours are a must. He is excellent at explaining the abstract concepts in the course. I think the course is well-paced, too, and as someone who struggles with exam-taking, the problem-set heavy grading is amazing and really makes me feel like my understanding of the material is reflected in the grading.
- Content is great fun! Professor is very organized, friendly, and effective.
- Interesting subject, good lecturer.
- The subject was covered in an extremely intuitive way and at a very good pace, allowing me to really get a good feel for it, the content was interesting, and lecture was given in a very engaging way
- Best class I ever took

For the Honors Analysis I course:

8 - What are the best aspects of this course?

Response Rate	11/20 (55%)

- Nice content, really good instructor.
- Great content, great instructor
- Very interesting and difficult homeworks, course content is cool.
- Fascinating subject matter, excellent lectures, interesting homeworks. A lot of additional resources provided by professor for students who are particularly interested in any given unit.
- Math.
- Very interesting topic. Engaging professor.
- The professor is very knowledgeable and very quick to answer questions about the content.
- Lectures were very helpful, as well as sections. I felt like I got a really good introduction to Analysis that touched on a lot of important concepts.
- Kobe is an excellent lecturer; he is engaging, smart, funny, and passionate. is incredibly organized, patient, knowledgable, and kind. Together, they make an incredible teaching team. It is difficult to provide concreteness to such abstract content, but I think that Kobe did an excellent job- providing examples and connecting to other areas of mathematics we may be familiar with. I think that I walked away with not only more knowledge of the branch of analysis, but also understanding of the larger mathematical structures that guide our study. Furthermore, weekly section with helped to solidify and review content and provide a very good starting point to tackle the homeworks; their office hours were also very helpful. Both and Kobe encouraged students to work together, which created a lovely classroom environment. Thank you both for a great semester!
- Kobe Marshall-Stevens is an excellent professor who is approachable and makes difficult concepts inviting.
- The content is just so peak. It opened my eyes to some weird beauty of mathematics that I just didn't think about before. The prof and TA are also very clearly engaged with the class and the learning, and it makes the experience great.

5.3 Anonymous feedback

In each Canvas course page I include a link to an anonymous Microsoft form with the following prompt:

"This form is for any feedback or concerns you have about the course that you wish to share anonymously. If you want a response or follow up, please include your email so I know who you are."

Below I include a sample of some responses I have received; I frequently remind students that they can (and should) provide this feedback. Some responses were positive and encouraging, while others provided constructive criticism; I have tried to include a representative sample.

Some positive examples included:

"Hi professor, I just wanted to say that you were very thorough and clear with your explanations today. I think you did a really good job teaching the chain rule. Keep up the great work :). Thanks!"

"I think today's class was very engaging because people were more interactive with the lecture!"

"I enjoy the welcoming atmosphere that allows students to feel comfortable asking questions!"

Some constructive criticisms included:

"It would be helpful if you wrote a bit more slowly and neater because at times it is difficult to copy everything down. I would also like less theory and more practical applications/examples."

"I feel like the class is allowing me to understand the content well, but the group projects and personal portfolio seem to be a lot of effort with minimal payoff as far as understanding."

"I was wondering if it would be possible to use section time to focus more on some of the proofs and less so the definitions, just since those are easier to review independently and (at least for me personally) it very much helps to hear the proofs multiple times."

For the constructive criticism, such as prompts to write bigger/neater and to slow down, I make sure to adopt this as the semester progresses. The responses about specific assignments and the use of section time informed me about how to alter the courses going forward; for instance, the second time I taught Calculus I we reduced the number of group projects and made sure to devote more section time to reviewing content from class.

6 Videos

During the Fall 2025 and Spring 2026 semesters I recorded all of the classes for the graduate sequence in partial differential equations. Playlists containing video recordings of the classes can be found below.

Partial Differential Equations I:

`https://www.youtube.com/playlist?list=PLbuZZ9RltZUKX5TtMOULQPBbIxwuF3rrN`

Partial Differential Equations II:

`https://www.youtube.com/playlist?list=PLbuZZ9RltZUKSgRingQor65Xg2yBMlZkX`

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